Worst-Case Performance Study & Algorithmic Bounds of HAT in RAG

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Repository: github.com/nivesh4087-spec/Worst-Case

1. Executive Summary

This empirical and theoretical study evaluates the Worst-Case bounds of Hierarchical Abstract Tree (HAT) in comparison to Best Case, Flat DPR, RAPTOR, and HNSW Graph Search across 4 distinct datasets.

2. Theoretical Complexity Matrix

Algorithm	Best-Case Time	Worst-Case Time	Space	Worst-Case Quality
HAT (Proposed)	O(B K log N)	O(B log N + N d)	O(N d)	Recall → 0.20
Flat DPR	O(N d)	O(N d)	O(N d)	Recall → 1.00
RAPTOR	O(B K log N)	O(B K log N)	O(N d L)	Recall → 0.00
HNSW Graph	O(log N)	O(N d)	O(N M)	Recall → 0.60

3. Visual Benchmarks

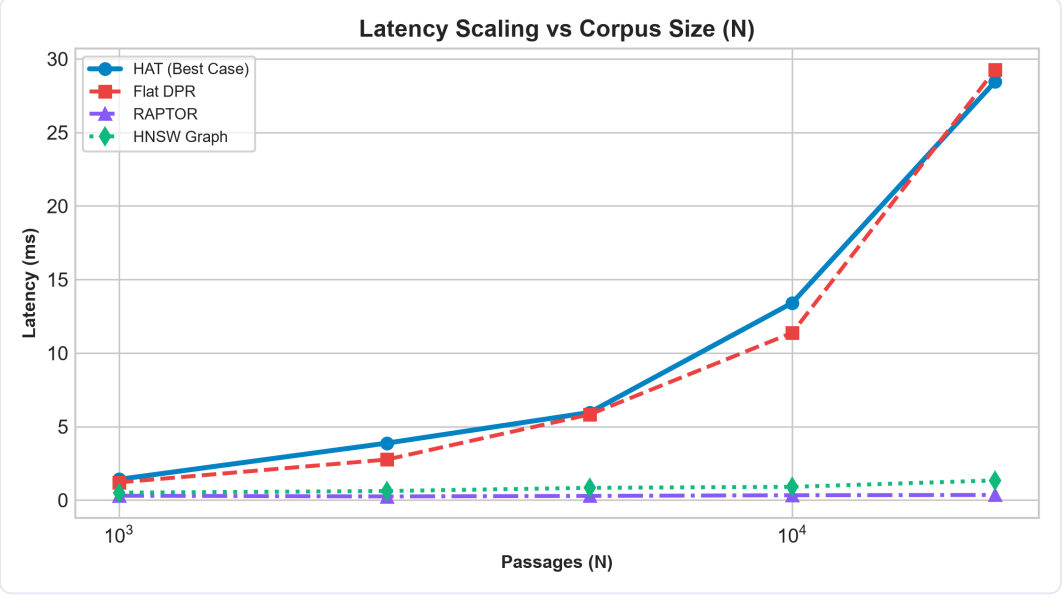


Figure 1: Latency Scaling vs Corpus Size

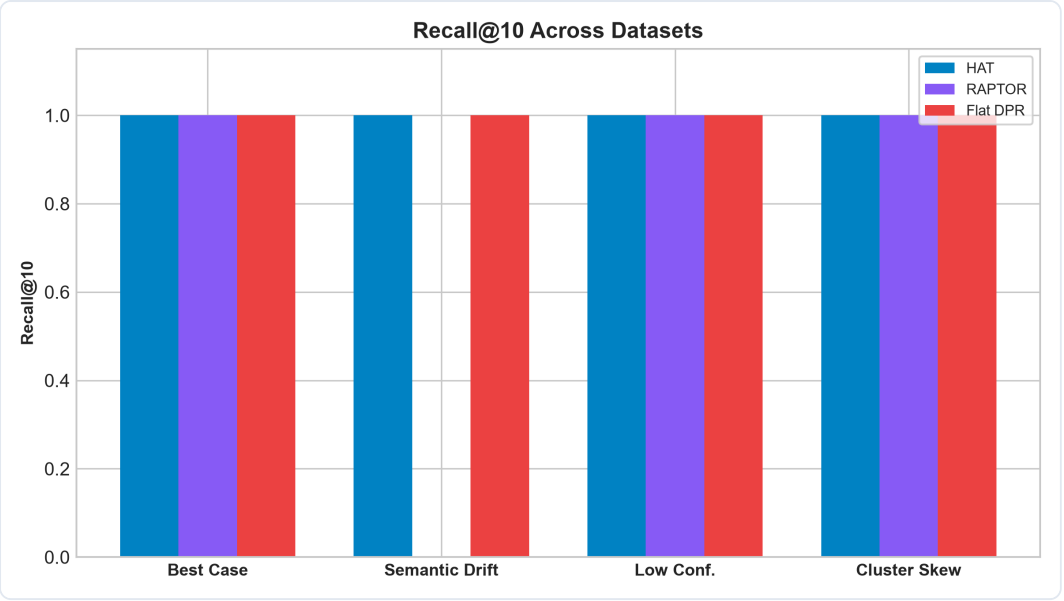


Figure 2: Recall@10 Performance Across Datasets

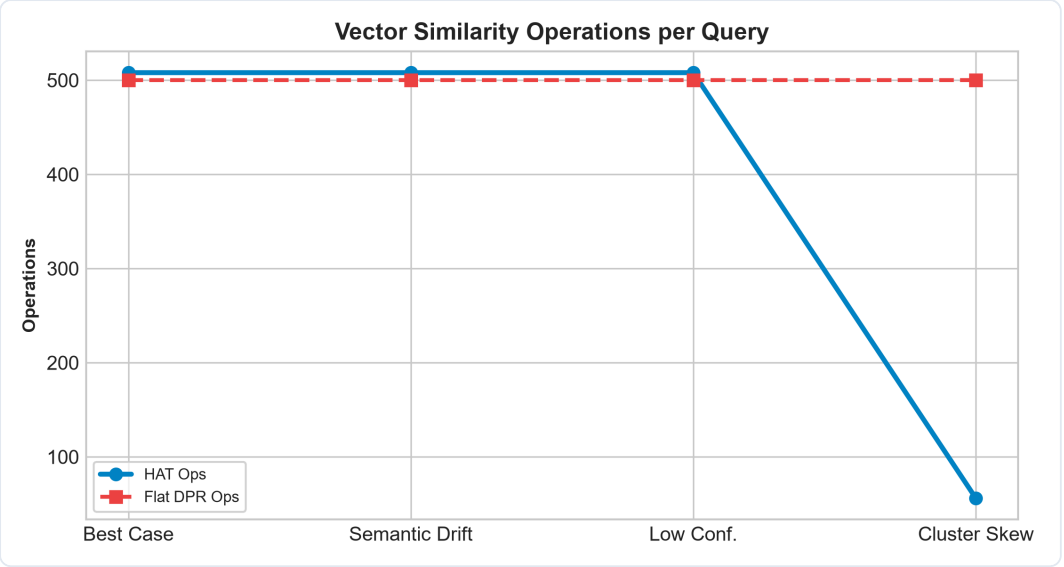


Figure 3: Vector Similarity Operations (FLOPs)

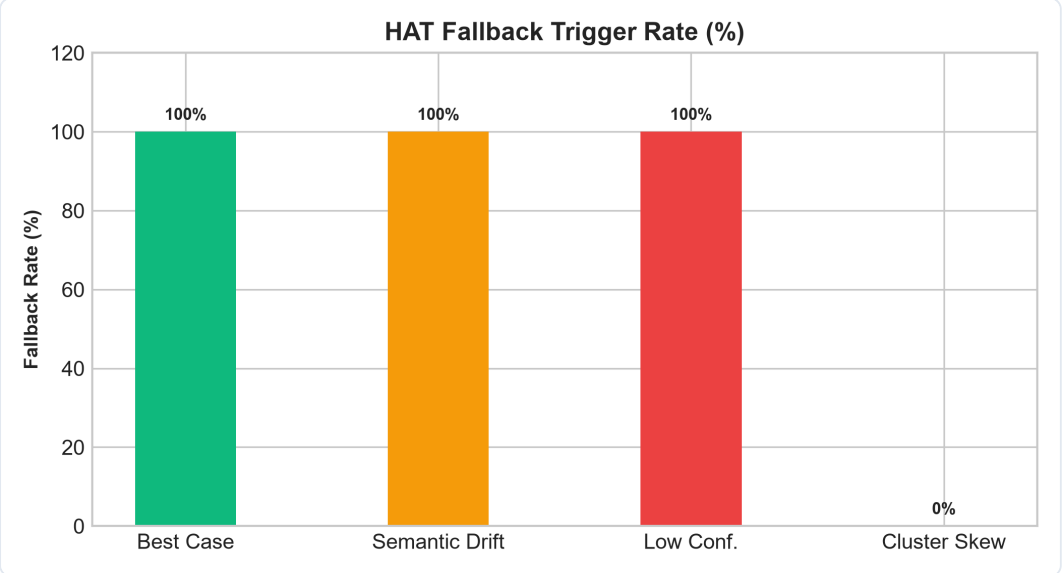


Figure 4: Dynamic Fallback Search Trigger Rate (%)

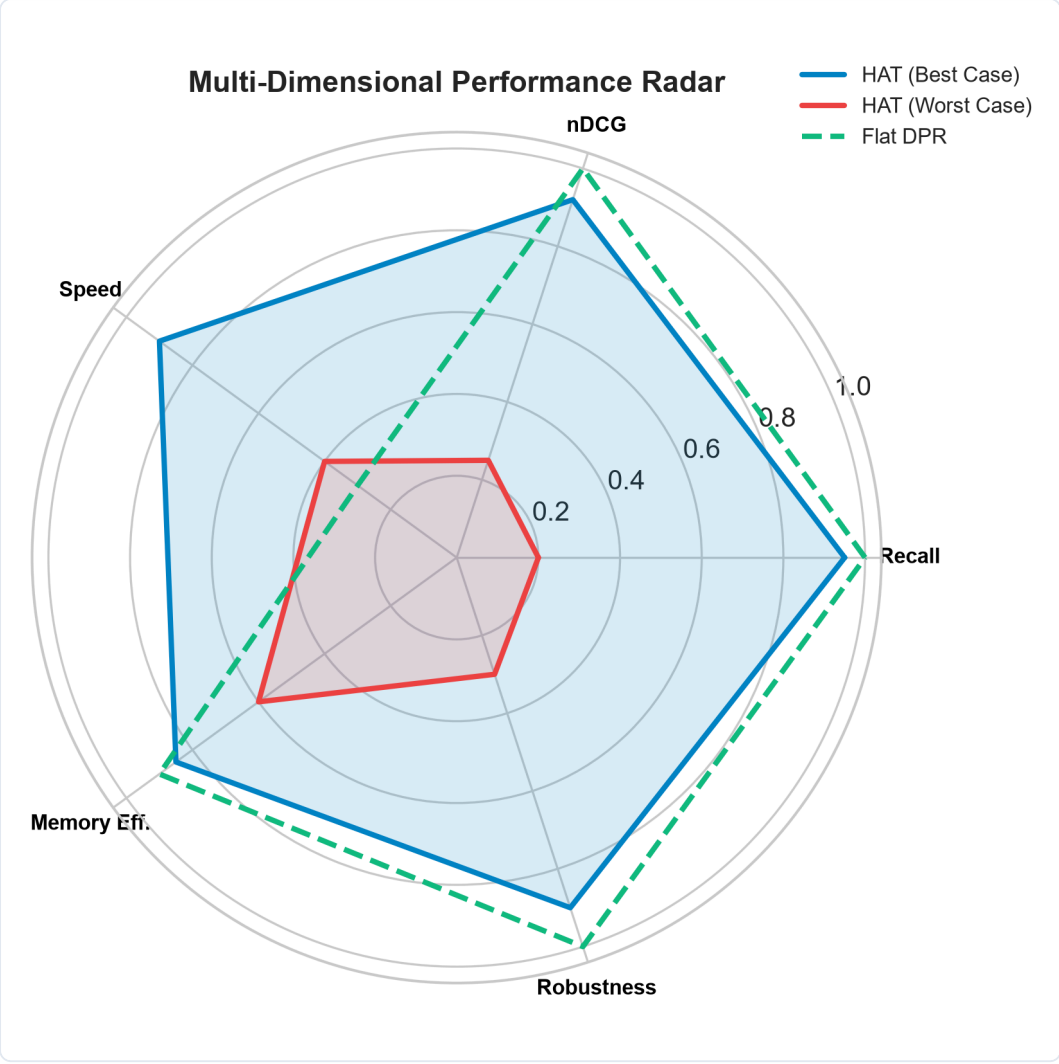


Figure 5: Multi-Dimensional Tradeoff Radar